

DAY 1 — PRACTICAL TASKS

Linux Networking & AWS EC2

Prepared by	Jayesh Patil
Organization	Fortune Cloud Technologies
Date	11 September 2026
Environment	Amazon Linux 2023 / AWS EC2
EC2 Private IPv4	172.31.16.110
EC2 Public IPv4	54.92.222.22

This report consolidates the submitted practical evidence into one PDF. Repeated screenshots were removed, and only screenshots that add distinct evidence are included.

Evidence note: The screenshots do not contain a complete before/after disconnect-reconnect record for Task 3, and they do not show an actual failed external access followed by a security-rule fix for Task 5. Those items are therefore marked as evidence gaps rather than being fabricated.

TASK 1 — Linux IP Investigation

Objective: Identify the Linux system IP address, active network interface, default gateway, DNS information, and overall network configuration.

Documented commands

```
ip a
ip route
cat /etc/resolv.conf
hostname -I
```

Observed values from the submitted terminal evidence

Item	Observed result
IPv4 address	172.31.16.110
Network interface	ens5
Subnet/prefix	172.31.16.0/20
Default gateway	172.31.16.1
DNS server	172.31.0.2

```
ec2-user@jayeshpatil:~  
[ec2-user@jayeshpatil ~]$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP  
    group default qlen 1000  
    link/ether 0a:ff:d8:85:da:21 brd ff:ff:ff:ff:ff:ff  
    altname enp0s5  
    altname eni-001822ae9b0ac9ab1  
    altname device-number-0.0  
    inet 172.31.16.110/20 metric 512 brd 172.31.31.255 scope global  
    dynamic ens5  
        valid_lft 3305sec preferred_lft 3305sec  
        inet6 fe80::8ff:d8ff:fe85:da21/64 scope link proto kernel_l1  
        valid_lft forever preferred_lft forever  
[ec2-user@jayeshpatil ~]$
```

Evidence 1 — `ip a`: active interface ens5 and IPv4 address 172.31.16.110.

```

ec2-user@jayeshpatil:~
[ec2-user@jayeshpatil ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP
    group default qlen 1000
    link/ether 0a:ff:d8:85:da:21 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-001822ae9b0ac9ab1
    altname device-number-0.0
    inet 172.31.16.110/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 3305sec preferred_lft 3305sec
        inet6 fe80::8ff:d8ff:fe85:da21/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
[ec2-user@jayeshpatil ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP
    mode DEFAULT group default qlen 1000
    link/ether 0a:ff:d8:85:da:21 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-001822ae9b0ac9ab1
    altname device-number-0.0
[ec2-user@jayeshpatil ~]$ ip route
default via 172.31.16.1 dev ens5 proto dhcp src 172.31.16.110 metric 512
172.31.0.2 via 172.31.16.1 dev ens5 proto dhcp src 172.31.16.110 metric 512
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.16.110 metric 512
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.16.110 metric 512

```

Evidence 2 — `ip route`: default route via 172.31.16.1 through ens5.

```
ec2-user@jayeshpatil:~  
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.16.110  
metric 512  
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.16.110 metri  
c 512  
[ec2-user@jayeshpatil ~]$ cat /etc/resolv.conf  
# This is /run/systemd/resolve/resolv.conf managed by man:systemd-  
resolved(8).  
# Do not edit.  
#  
# This file might be symlinked as /etc/resolv.conf. If you're look  
ing at  
# /etc/resolv.conf and seeing this text, you have followed the sym  
link.  
#  
# This is a dynamic resolv.conf file for connecting local clients  
directly to  
# all known uplink DNS servers. This file lists all configured sea  
rch domains.  
#  
# Third party programs should typically not access this file direc  
tly, but only  
# through the symlink at /etc/resolv.conf. To manage man:resolv.co  
nf(5) in a  
# different way, replace this symlink by a static file or a differ  
ent symlink.  
#  
# See man:systemd-resolved.service(8) for details about the suppor  
ted modes of  
# operation for /etc/resolv.conf.  
  
nameserver 172.31.0.2  
search ec2.internal  
[ec2-user@jayeshpatil ~]$ hostname -I  
172.31.16.110  
[ec2-user@jayeshpatil ~]$
```

Evidence 3 — `/etc/resolv.conf`: DNS server 172.31.0.2 and search domain ec2.internal.

TASK 2 — IPv4 Address Analysis

Objective: Analyze the assigned IPv4 address, identify its four octets, determine private/public classification, and show reachable systems.

IPv4 analysis

Assigned IPv4: **172.31.16.110**

Octet	Value
1	172
2	31
3	16
4	110

Classification: 172.31.16.110 is a **private IPv4 address** because it falls within the 172.16.0.0–172.31.255.255 private range.

Reachability evidence: The submitted terminal shows successful ICMP connectivity to 8.8.8.8 and google.com. These are reachable external systems, not local LAN devices.

```
ec2-user@jayeshpatil:~  
[ec2-user@jayeshpatil ~]$ ip -4 addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue  
state UNKNOWN group default qlen 1000  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qd  
isc mq state UP group default qlen 1000  
    altname enp0s5  
    altname eni-001822ae9b0ac9ab1  
    altname device-number-0.0  
    inet 172.31.16.110/20 metric 512 brd 172.31.31.255  
    scope global dynamic ens5  
        valid_lft 2974sec preferred_lft 2974sec  
[ec2-user@jayeshpatil ~]$ hostname  
jayeshpatil  
[ec2-user@jayeshpatil ~]$ hostname -I  
172.31.16.110  
[ec2-user@jayeshpatil ~]$ ip -4 route  
default via 172.31.16.1 dev ens5 proto dhcp src 172.31  
.16.110 metric 512  
172.31.0.2 via 172.31.16.1 dev ens5 proto dhcp src 172  
.31.16.110 metric 512  
172.31.16.0/20 dev ens5 proto kernel scope link src 17  
2.31.16.110 metric 512  
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.  
16.110 metric 512  
[ec2-user@jayeshpatil ~]$ curl ifconfig.me  
54.92.222.22[ec2-user@jayeshpatil ~]$ curl ifconfig.me  
54.92.222.22[ec2-user@jayeshpatil ~]$ ping -c 4 192.16  
ping -c 4 192.168.1.1  
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
```

Evidence 4 — IPv4 address, hostname, route table, and connectivity attempt shown together.

```

ec2-user@jayeshpatil:~
[ec2-user@jayeshpatil ~]$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    altname enp0s5
    altname eni-001822ae9b0ac9ab1
    altname device-number-0.0
    inet 172.31.16.110/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 3429sec preferred_lft 3429sec
[ec2-user@jayeshpatil ~]$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=1.05 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=1.05 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=116 time=1.07 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=116 time=1.07 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 1.048/1.060/1.073/0.010 ms
[ec2-user@jayeshpatil ~]$ ping -c 4 google.com
PING google.com (172.253.115.101) 56(84) bytes of data.
64 bytes from bg-in-f101.1e100.net (172.253.115.101): icmp_seq=1 ttl=106 time=1.49 ms
64 bytes from bg-in-f101.1e100.net (172.253.115.101): icmp_seq=2 ttl=106 time=1.55 ms
64 bytes from bg-in-f101.1e100.net (172.253.115.101): icmp_seq=3 ttl=106 time=1.54 ms
64 bytes from bg-in-f101.1e100.net (172.253.115.101): icmp_seq=4 ttl=106 time=1.59 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.488/1.542/1.589/0.035 ms
[ec2-user@jayeshpatil ~]$

```

Evidence 5 — successful ping to 8.8.8.8 and successful DNS resolution/ping of google.com (4/4 replies).

TASK 3 — Dynamic IP Investigation

Objective: Record the current IP, check connection status, disconnect/reconnect when safe, and compare the address before and after.

Evidence status: INCOMPLETE IN THE SUBMITTED SCREENSHOTS

The uploaded screenshots show a dynamically assigned private address (the `ip` output includes the word **dynamic** for ens5), but they do not contain the required controlled disconnect → reconnect sequence.

What is evidenced

Current private IPv4: **172.31.16.110**

Interface: **ens5**

Address state: **dynamic**

Required evidence still to capture before final submission

1. Record IP before disconnect.
2. Record network connection status.
3. Safely disconnect and reconnect the network (only on a local/safe environment; do not do this over an SSH session).
4. Record IP after reconnect.
5. Show a terminal record: **Before disconnect** → **IP Address** and **After reconnect** → **IP Address**.

TASK 4 — Cloud Linux Server & IP

Objective: Create/access an AWS EC2 Linux server, connect using SSH, find private/public IP information, and compare AWS console with Linux.

Documented commands

```
ssh -i linux.pem ec2-user@54.92.222.22
hostname -I
ip -4 addr
ip route
curl -4 ifconfig.me
```

Observed EC2 addressing

Source	Private IPv4	Public IPv4
AWS EC2 Console	172.31.16.110	54.92.222.22
Linux terminal	172.31.16.110	54.92.222.22

Result: The private IP shown in the AWS console matches the private IP shown inside Linux. The public IP 54.92.222.22 is the address used to reach the instance from the internet.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) jayeshP (871623373272) jayeshP

EC2 > Instances

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- ▼ Images
 - AMIs
 - AMI Catalog
- ▼ Elastic Block Store

Instances (1/9)

Last updated about 8 hours ago

Connect Instance state Actions Launch instances

Saved filter sets: Choose filter set

Filter Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☒	volume2	i-099754dfd4215f2c6	Running	t3.micro	Initializing	us-east-1c
☐	volume	i-03f82658b3fb0dabf	Stopped	t3.micro	—	us-east-1c
☐	portfolio	i-0f917b3b04744b374	Stopped	t3.micro	—	us-east-1a

i-099754dfd4215f2c6 (volume2)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary

Instance ID: i-099754dfd4215f2c6

Public IPv4 address: 54.92.222.22 | open address

Private IPv4 addresses: 172.31.16.110

Public DNS

CloudShell Agent Toolkit for AWS Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Evidence 6 — AWS EC2 console showing instance volume2 with Public IPv4 54.92.222.22 and Private IPv4 172.31.16.110.

```
ec2-user@jayeshpatil:~  
Admin@jaypc MINGW64 ~/Downloads  
$ ssh -i linux.pem ec2-user@54.92.222.22  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
  
#_      _____ Amazon Linux 2023  
'\    #####_\n~~   '\#####\  
~~   \####|\n~~   \#/ \n~~   V~' '-> https://aws.amazon.com/linux/amazon-linux-2023  
~~~~  
~~~~_.-._\n     _/_/\n     /m/'  
  
Last login: Fri Sep 11 11:42:03 2026 from 103.252.53.70  
[ec2-user@jayeshpatil ~]$ curl -4 ifconfig.me  
54.92.222.22 [ec2-user@jayeshpatil ~]$
```

Evidence 7 — successful SSH login to Amazon Linux 2023 followed by `curl -4 ifconfig.me`, returning 54.92.222.22.

TASK 5 — Cloud Network Troubleshooting

Objective: Systematically check IP, interface, route, connectivity, services, listening ports, and AWS security rules; identify and fix the access problem.

Documented troubleshooting commands

```
sudo systemctl status nginx
ip -4 addr
ip route
ping -c 4 8.8.8.8
sudo ss -tulpn
curl http://localhost
```

Observed results

Check	Observed result
IP address	172.31.16.110 on ens5
Interface	ens5 is UP
Default route	172.31.16.1 via ens5
Connectivity	8.8.8.8: 4/4 replies; google.com: 4/4 replies
Nginx service	active (running)
Listening port	TCP port 80 is listening
Local HTTP test	curl http://localhost returns the Nginx welcome page
AWS inbound rule	Submitted console screenshot shows TCP 80 allowed from 0.0.0.0/0

Important evidence limitation: The submitted AWS screenshot already shows TCP/80 allowed. Therefore, the screenshots do not support claiming that a Security Group rule caused a failure or that a rule was changed as the fix. The evidence does support that Nginx was running, port 80 was listening, and localhost HTTP worked.

```

ec2-user@jayeshpatil:~$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-09-11 11:51:03 UTC; 9s ago
     Process: 3778 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0)
     Process: 3779 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 3780 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
  Main PID: 3781 (nginx)
    Tasks: 3 (limit: 1059)
   Memory: 3.6M
      CPU: 58ms
   CGroup: /system.slice/nginx.service
           └─3781 "nginx: master process /usr/sbin/nginx"
             └─3782 "nginx: worker process"
               └─3783 "nginx: worker process"

Sep 11 11:51:03 jayeshpatil systemd[1]: Starting nginx.service - The nginx HTTP and
Sep 11 11:51:03 jayeshpatil nginx[3779]: nginx: the configuration file /etc/nginx/n
Sep 11 11:51:03 jayeshpatil nginx[3779]: nginx: configuration file /etc/nginx/nginx
Sep 11 11:51:03 jayeshpatil systemd[1]: Started nginx.service - The nginx HTTP and

[ec2-user@jayeshpatil ~]$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default ql
en 1000
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default
qlen 1000
    altname enp0s5
    altname eni-001822ae9b0ac9ab1
    altname device-number-0.0
    inet 172.31.16.110/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 3429sec preferred_lft 3429sec
[ec2-user@jayeshpatil ~]$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=1.05 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=1.05 ms

```

Evidence 8 — Nginx service is active (running), with the server IP and connectivity commands visible below.

```

ec2-user@jayeshpatil:~
[ec2-user@jayeshpatil ~]$ sudo ss -tulpn
Netid State Recv-Q Send-Q           Local Address:
Port                Peer Address:Port
Process
udp    UNCONN 0      0      0.0.0.0:*      127.0.0.1:
323    users: (("chronyd",pid=1896,fd=5))
udp    UNCONN 0      0      172.31.16.110%ens5:
68     0.0.0.0:*
users: (("systemd-network",pid=1708,fd=17))
udp    UNCONN 0      0      [::]:*        [::]:1:
323    [::]:*
users: (("chronyd",pid=1896,fd=6))
udp    UNCONN 0      0      [fe80::8ff:d8ff:fe85:da21]%ens5:
546    [::]:*
users: (("systemd-network",pid=1708,fd=19))
tcp    LISTEN 0      511     0.0.0.0:
80     0.0.0.0:*
users: (("nginx",pid=3783,fd=8),("nginx",pid=3782,fd=8),("ngin
x",pid=3781,fd=8))
tcp    LISTEN 0      128     0.0.0.0:
22     0.0.0.0:*
users: (("sshd",pid=1868,fd=7))
tcp    LISTEN 0      511     [::]:*        [::]:
80     [::]:*
users: (("nginx",pid=3783,fd=9),("nginx",pid=3782,fd=9),("ngin
x",pid=3781,fd=9))
tcp    LISTEN 0      128     [::]:*        [::]:
22     [::]:*
users: (("sshd",pid=1868,fd=8))
[ec2-user@jayeshpatil ~]$
[ec2-user@jayeshpatil ~]$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and w
orking.
Further configuration is required for the web server, reverse
proxy,
API gateway, load balancer, content cache, or other features.<

```

Evidence 9 — `ss -tulpn` shows TCP port 80 listening for nginx, and `curl http://localhost` returns the Nginx page.

FINAL SUBMISSION CHECKLIST

Use this page as the final review before uploading the PDF. The assignment asks for all five practical tasks, screenshots of major steps, commands and outputs, and all task screenshots in one PDF. ■filecite■turn0file0■L5-L9■

Task	Evidence included	Status
Task 1 — Linux IP Investigation	IP, interface, route, DNS screenshots + documented values	Evidenced
Task 2 — IPv4 Analysis	IPv4/octets + reachable external systems	Evidenced
Task 3 — Dynamic IP	Current dynamic IP shown, but no disconnect/reconnect before-after record	Needs screenshot
Task 4 — EC2 & IP	AWS console + SSH + public/private IP evidence	Evidenced
Task 5 — Troubleshooting	IP/interface/route/connectivity/service/port/security evidence	Needs failure→fix evidence

Recommended filename: Day1_Tasks_Submission_Jayesh_Patil.pdf

Before submission: Capture the missing Task 3 before/after terminal record and, if the trainer expects a real troubleshooting incident, capture a genuine Task 5 failure and its subsequent fix. Do not add invented screenshots or results.